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Q6 ⇒ Verify $A\,(\text{Adj}\,A) = |A|\,I$, where $A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$

$A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix} \cdot \text{Adj}\,A \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$

Sol

$$\begin{bmatrix} 2\times(-6)+(3)(4) & 2\times(-3)+3(2) \\ -4\times(-6)+(-6)(4) & (-4)(-3)\times -6(2) \end{bmatrix}$$

$$\begin{bmatrix} \cancel{-12+12} & \cancel{-6+6} \\ 24-24 & \cancel{+12-12} \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A\,(\text{Adj}) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

(Now)

$$|A| = \begin{vmatrix} 2 & 3 \\ -4 & -6 \end{vmatrix}$$

$-12+12=0$

$|A| = 0$

$$0 \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Hence prove.

Q7, Q no 7
Find the inverse of the following matrices by adjoint method if exist.

(i) $A = \begin{bmatrix} 3 & 2 \\ -2 & -1 \end{bmatrix}$

$A^{-1} = \frac{Adj(A)}{|A|}$

$Adj(A) = \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix}$

$|A| = \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix}$

$-3 + 4$

$|A| = 1$

$A^{-1} = \frac{Adj(A)}{|A|}$

$A^{-1} = \frac{\begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix}}{1}$

$A^{-1} = \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix}$

Q8/ Find the solution of Matrix by MatRix inversion method and crammer rule.

$$2x + 3y = 14$$
$$-4x + y = 28$$

Sol/

$$\begin{bmatrix} 2 & 3 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 14 \\ 28 \end{bmatrix}$$

$$A \quad X = B$$

Now we have to find inverse

$$X = A^{-1} B \rightarrow eq①$$

Note $A^{-1} = \frac{1}{|A|} adj(A)$

$|A| = 14 \neq 0$ solution exist

$$Adj\,A = \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{14} \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix}$$

Now

$$X = A^{-1} B$$

$$X = \begin{bmatrix} 1 & -3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 14 \\ 28 \end{bmatrix}$$

$$X = \begin{bmatrix} 14 \times 1 + (-3)\,28 \\ 4 \times 14 + (2)\,28 \end{bmatrix} \times \frac{1}{14} =$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{14} \begin{bmatrix} -70 \\ 112 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -5 \\ 8 \end{bmatrix}$$

hence

$$x = -5$$
$$y = 8$$